

Haonan Zhao

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OBJECTIVE

Seeking a PhD/MPhil position in 27 Fall or 28 Spring. My current research focuses on MLLMs/diffusion MLLMs, with an emphasis on reasoning and hallucination. My prior research experience centered on computer vision.

EDUCATION

- **University of Auckalnd** Jul 2026 - Jun 2027 (Expected)
Bachelor of Science in Mathematics (joint program with NEU) New Zealand
- **Northeastern University** Sep 2023 - Jun 2026
Bachelor of Science in Mathematics and Applied Mathematics China
 - GPA: 89.86/100.0, Rank: 3/105

HONORS & AWARDS

- **International Student Excellence Scholarship** Jun 2026
University of Auckland | Selected as one of 50 recipients 
- **Academic Excellence Scholarship** 2023 - 2025
Northeastern University

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=IN SUBMISSION

- [S.2] **H. Zhao, Z. Qiu, Y. Fu, X. Luo, and X. Zhang. Look Before You Denoise: Understanding and Calibrating Visual Attention Sinks for Reliable Reasoning in dMLLMs.** Submitted to *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2027, **under review**.
- [S.1] **S. Wang, X. Wang, H. Zhao, J. Xu, B. Jiang, L. Zhu, X. Zhao, Y. Tian, and J. Tang. Decoupling Amplitude and Phase Attention in Frequency Domain for RGB-Event Based Visual Object Tracking.** Submitted to *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)*, 2026, **under review**.
- [C.2] **H. Zhao, W. Pan, R. Xie, Z. Qiu, Y. Ma, and R. Luo. Learning Where to Attend: A Sparse Dynamic Gate Transformer for Efficient Image Deraining.** *IEEE International Conference on Multimedia and Expo (ICME)*, **Spotlight**, 2026.
- [C.1] **Y. Song, Y. Ma, Z. Chu, H. Zhao, R. Luo, and R. Su. GMDNet: A Graph-Dynamics-Aware Multi-Scale Decoupling Network for Spatio-Temporal Traffic Flow Prediction.** *IEEE International Conference on Intelligent Transportation Systems (ITSC)*, 2025.
- [J.1] **L. Li, H. Zhao, L. Lyu, and F. Yang. Multi-strategy Improved Gazelle Optimization Algorithm for Numerical Optimization and UAV Path Planning.** *Scientific Reports*, 2025.

RESEARCH EXPERIENCE

- **University of Wisconsin–Madison** Apr 2026 - Present
Research Intern, supervised by Prof. Xiao Luo Remote
 - Analyzed visual attention sinks in diffusion MLLMs, revealing their progressive evolution across layers.
 - Designed a training-free method, named Attention Residual Calibration, to improve visual grounding.
 - Achieved consistent improvements across multiple hallucination and multimodal benchmarks.
- **Nanyang Technological University** May 2025 - Jan 2026
Research Intern, supervised by Dr. Ruikang Luo and Prof. Rong Su Remote
 - Developed a sparse Transformer for deraining with content-adaptive gating and frequency-domain learning.
 - Achieved sota performance, outperforming 12 baselines by 0.58 dB PSNR with strong cross-task generalization.
 - Co-authored the related work for GMDNet and conducted benchmark experiments on PEMS04 and PEMS08.
- **Anhui Provincial Key Lab of Multimodal Cognitive Computation** Jun 2025 - Sep 2025
Research Intern, supervised by Prof. Xiao Wang On site
 - Co-authored the related work, experimental visualization, and attribute analysis for APMTrack.
 - Evaluated APMTrack on 12 COESOT attributes, achieving the highest success rate on 11 attributes.
 - Developed Python scripts for result visualization and comparative analysis.

INDUSTRIAL EXPERIENCE

- **IFLYTEK CO.LTD** Jul 2024 - Sep 2024
Resource Engineer Intern Wuhu, China
 - Collected and organized text, audio, and video data for LLM datasets.
 - Annotated satellite and equipment-monitoring images using LabelImg and identified resource distribution regions.
 - Builded an object detection dataset with 98% annotation accuracy.

PROJECT

- **UoA Maths 399 Capstone: Symmetric Chaos** Aug 2026 - Present
Team Member, supervised by Prof. Claire Postlethwaite
 - Implemented MATLAB algorithms to generate and visualize symmetric chaotic patterns.
 - Analyzed the symmetries and dynamical behavior of the resulting chaotic systems.
 - Investigated parameter effects, numerical stability, and computational efficiency.

SKILLS

- **Programming Languages:** Python, C/C++, R, MATLAB
- **Development & Research Tools:** Git, Linux, LaTeX, Codex
- **ML & Data Analysis:** PyTorch, Transformers, scikit-learn, Time Series Analysis, PCA
- **Languages:** Mandarin (Native), English (Fluent)

RELEVANT COURSEWORK

- **Statistics & Optimization:** Probability Theory, Mathematical Statistics, Machine Learning, Operations Research
- **Mathematics:** Matrix Analysis, Algebraic Structures, Combinatorics, Numerical Computation, ODE, PDE
- **Computer Science:** C++ Programming, Computational Techniques and Computer Systems